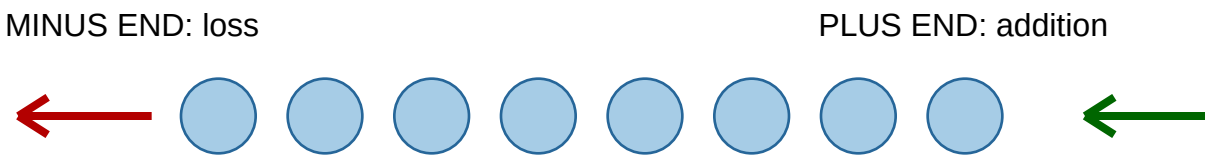
Filament treadmilling

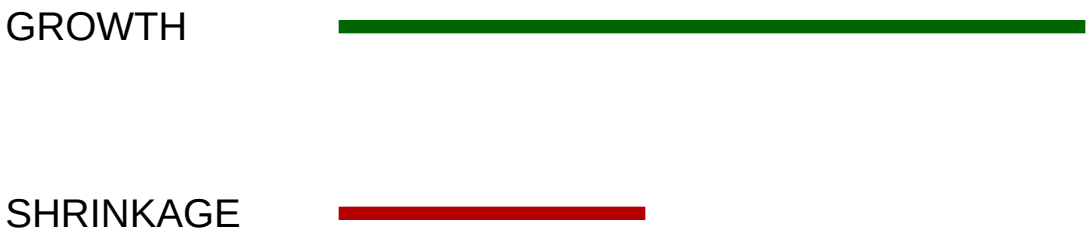
Treadmilling is a behavior of actin filaments in which addition of subunits at the plus end balances loss at the minus end. Equal addition and loss maintain near-constant length. Greater addition than loss causes net growth. The diagram shows balanced turnover, not whole-filament translation.



Equal addition and loss: length stays approximately constant.

Microtubule dynamic instability

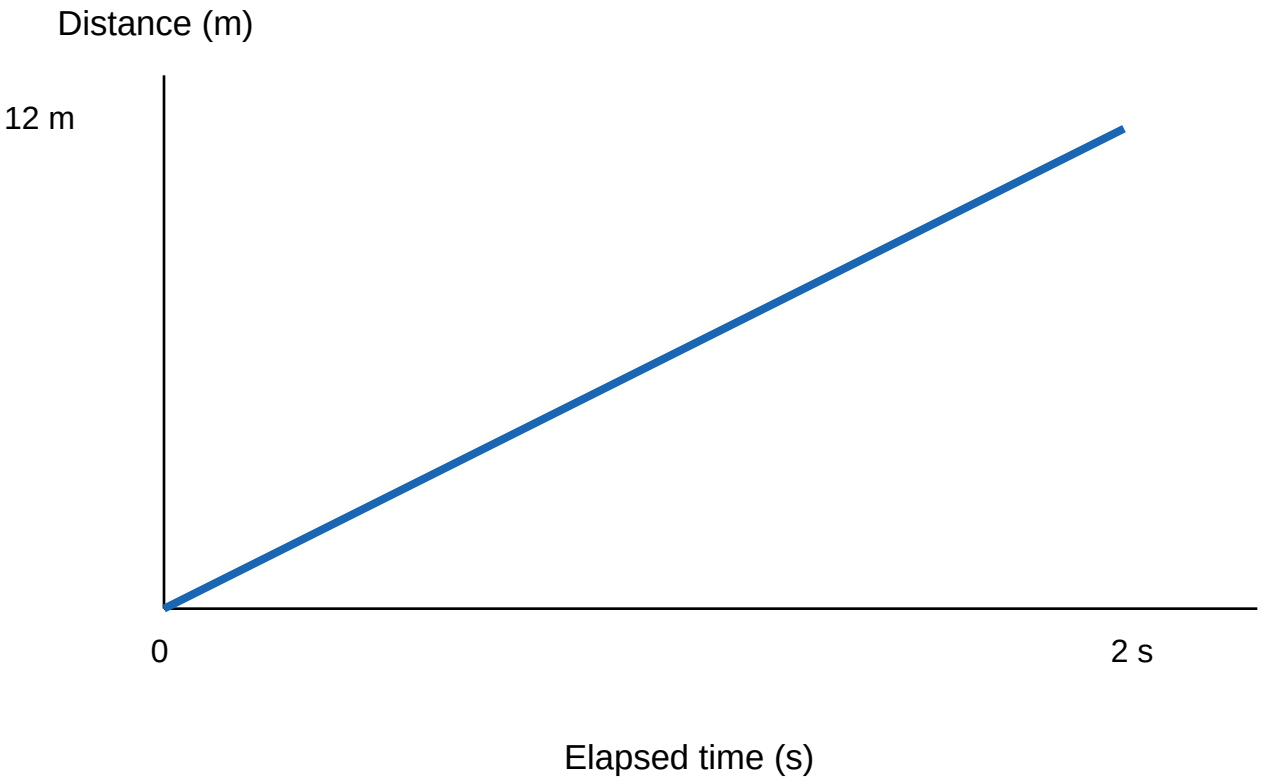
Dynamic instability is switching between microtubule growth and shrinkage. It differs from treadmilling, which describes balanced turnover at near-constant length. The diagram shows two alternate states, not simultaneous balanced turnover.



Alternate states: growth <-> shrinkage

Speed from distance and time

Speed equals distance divided by elapsed time: v = d/t. Distance d is in metres (m), elapsed time t is in seconds (s), and speed v is in metres per second (m/s). Example: d = 12 m and t = 2 s give v = 6 m/s.



Force, mass and acceleration

Force equals mass multiplied by acceleration: F = m*a. Mass m is in kilograms (kg), acceleration a is in metres per second squared (m/s^2), and force F is in newtons (N). Example: m = 2 kg and a = 3 m/s^2 give F = 6 N.

$$F = m * a$$

$$2\ \text{kg} * 3\ \text{m/s}^2 = 6\ \text{N}$$